

Combining CNN with Autoencoder for Motor-Imagery Classification

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Abstract— Motor Imagery (MI) is a significant component to start with Brain-Computer-Interface (BCI) research. For the past decade, EEG (Electroencephalogram) signal-based MI classification remains a major area of interest for researchers. In this paper, the accuracy of the multi-class classification of MI with EEG signal using machine learning (One-vs-One and One-vs-Rest), and Deep Learning (CNN - Convolutional Neural Network) using pre-processing techniques such as Wavelet Transform (WT) and Short-Time Fourier Transform (STFT) has been compared. The EEG signals are acquired from the PhysioNet Database (EEG Motor Movement/Imagery Dataset Version 1.0). From the comparative analysis study, the Deep Learning approach for Wavelet Transformed EEG signal gave the highest overall accuracy of 96.68% and AUC score of 0.87.

Keywords— EEG signal, Motor Imagery, One-vs-One, One-vs-Rest, Convolutional Neural Network, Wavelet Transform, Short-Time Fourier Transform.

I. INTRODUCTION

The brain is considered to be the most complicated organ of the body because of the complex connections between neurons [1]. EEG is an important tool that can be used to analyse brain function. EEG signals are recorded using the scalp electrodes [2][3][4]. MI is an intellectual process where the subject under consideration imagines moving limbs without actually moving limbs. MI is an important area of interest for researchers who are into developing cognitive tools or assistive devices for the subjects who are recovering from injury or motor disability [5][6]. MI tasks produce a similar EEG pattern as a real movement of limbs [7].

Event-Related Synchronization (ERS) and Event-Related Desynchronization (ERD) are the two phenomena that occur whenever the subject either imagines or performs the real movement of the limbs [8]. The ERS/ERD pattern is produced in the primary motor and sensory areas of the brain. EEG is considered a convenient tool to record the MI-EEG signals and further used for BCI applications [9]. The BCI applications also include the robotic-arm movement. MI classification in recent trends has become a major area of research, many attempt has been made by the researchers to find an optimal method to classify MI tasks [10].

The outline of major contribution is given as follows:

- Comparative analysis of ML and DL approach in MI classification.

- The feature learning is performed using Stacked Autoencoder (SAE) followed by CNN for the classification of MI tasks.

The organization of the paper is as follows. Section II includes the existing methods in the relation to the proposed method. Section III consists of the proposed methodology with a diagrammatic representation of the workflow. Section IV describes the result and discussion of different classification approaches. Section V describes the overall conclusion.

II. RELATED WORK

P. Yang et al. [11] proposed a novel method for feature extraction of EEG signals called Multiple-Riemannian-Covariance (MRC) - Multiple-Layer-Perceptron (MLP). This model is said to be fully connected with enhanced loss function and it was said that the proposed method for motor imagery classification shows better results when compared with that of the existing method. The mean accuracy of the model is claimed to be 76%. Sai Ho Ling et al. [12] proposed a comparison study of EEG motor imagery binary class classification using two types of datasets namely layered and stacked. Continuous Wavelet Transforms (CWT) to transform EEG signals into a scaleogram. The sequence of steps followed to achieve the main objective starts with preprocessing (Denoising) of EEG signals followed by transforming EEG signals in the frequency range from 8-32 HZ using CWT and finally, classification using the CNN architecture. They claim that the layered dataset has an accuracy of 68.33% and it is the highest accuracy obtained among the two types of the dataset used.

Aimin Jiang et al. [13] proposed a filtering technique named Spatio-Temporal filtering to attenuate the impact of noisy EEG channels and to validate the proposed filtering technique EEG motor imagery dataset has been used. Mustafa Alper Balim and Nurettin Acir [14] used SAE framework to classify the motor imagery limb movements. They claimed the accuracy to be 80.44% for the BCI IV 2b dataset. Biao Sun et al. [15] proposed a novel method Sparse spectro-temporal decomposition - Squeeze and Excitation - Convolutional Neural Network (SSD-SE-CNN) for EEG motor imagery classification. SSD algorithm is used for feature extraction, SE to recalibrate the feature response of each EEG channel, and CNN for classification. It was claimed by the authors that the proposed method outperforms the existing methods for classification. The SSD-SE-CNN method is said to have high accuracy, high robustness, and high efficiency. Tianwei Shi et al. [16]

proposed a novel method for feature extraction i.e. Common Spatial Pattern - Adaptive Auto-Regressive (CSP-ARR). The extracted features are fed into a Bayesian, linear discriminant, and common space classifier for comparison. Finally, it was claimed by the authors that the proposed method can be used as an efficient feature extraction technique in the case of EEG motor imagery classification.

Xuelin Ma et al. [17] proposed a method named Deep-Channel-Correlation-Network where the correlation between the EEG channels was considered as the features. This method could be used to classify 3-class motor imagery tasks of the same limb. The authors claim the accuracy for the proposed method is 87.03%. Yang Qiu et al. [18] proposed a system for automatic seizure detection using denoising sparse autoencoder and logistic regression. This method was used to classify EEG signals as epileptic or non-epileptic. The average sensitivity, recognition, and specificity for the proposed method are found to be 100%. Roya Arabshahi et al. [19] evaluated the performance of classification of EEG binary class MI task using CNN and SAE in combination to reduce the dimension of the data. It was claimed by the authors that the accuracy is found to be 82%.

III. PROPOSED METHODOLOGY

In this work, the EEG signals are pre-processed using WT and STFT separately. The autoencoder model is used for feature learning from the pre-processed data and finally the Multi-class classification approach to classify the MI tasks. The combination of deep learning and machine learning technique has been used to improve the accuracy of classification. This work aims to give a comparative analysis between Machine Learning (ML) approaches like One-vs-One (OvO) and One-vs-Rest (OvR) using Support Vector Machine (SVM) and Deep Learning (DL) approach using CNN.

This section describes the MI dataset used for this work, and the processes used in the proposed methodologies, namely pre-processing, feature extraction, and Multi-class classification using ML and DL approaches. Fig 1 gives the workflow of the proposed methodology. To compare the performance of ML and DL approach in MI classification, the different cases in the experiment, which have been considered, are as follows:

- *Case 1: ML and DL techniques performance using WT.*
- *Case 2: ML and DL techniques performance using STFT.*

A. Dataset description

The EEG MI Dataset for this work has been collected from PhysioNet Database [20]. The dataset comprises EEG recordings from 109 subjects. The data has been acquired using 64 electrodes (excluding A1, A2, F9, F10, FT10, FT9, Nz, P9, P10, TP10, and TP9) stored in EDF (European Data Format) file format and every signal is sampled with the sampling frequency of 160Hz. Every subject was asked to carry out 14 trials i.e. 2 one-minute

trials (Baseline run with eyes opened and eyes closed) followed by 3 two-minute trials for the below-given tasks:

- Either right or left fist - opening & closing.
- Imagining right or left fist - opening & closing.
- Either both fists or both feet - opening & closing.
- Imagining either both fists or both feet - opening & closing.

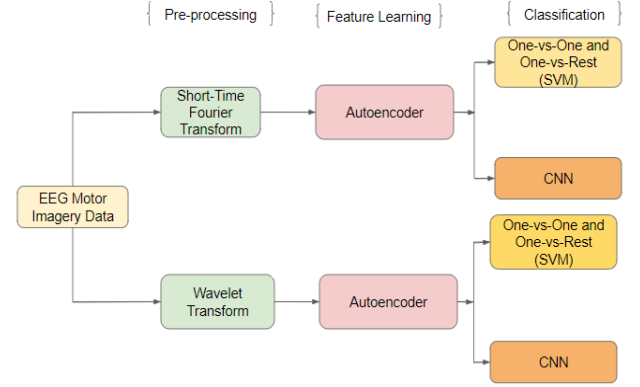


Fig. 1. Workflow of the proposed methodology.

Every annotation file includes three codes i.e. T0 (denotes rest state), T1 (denotes movement of either left fist or both the fists), and T2.

B. Pre-processing

EEG signals are non-stationary in nature; hence time-frequency analysis methods like Fourier Transform, STFT, WT, Stockwell Transform, etc. will be required to analyze the signals. In this work, Wavelet Transform and Short-Time Fourier Transform have been used to perform EEG signal analysis. In this application WT and STFT has been used for pre-processing because they provide better time and frequency resolution when compared to other signal transformation methods.

1) Short-Time Fourier Transform

STFT can be used to analyze EEG signals that are considered non-stationary in nature with multi-component. STFT is performed using a fixed window for analysis. The following are the types of STFT:

a) Continuous-time STFT

$$X(\tau, w) = \int_{-\infty}^{+\infty} x(t) \omega(t - \tau) e^{-iwt} dt \quad (1)$$

Where, $\omega(\tau) \rightarrow$ window function, $w \rightarrow$ frequency, $x(t) \rightarrow$ given signal, $X(\tau, w) \rightarrow$ transformed signal, $t \rightarrow$ time.

b) Discrete-time STFT

$$X(m, w) = \sum_{n=-\infty}^{+\infty} x[n] \omega[n - m] e^{-jwn} \quad (2)$$

Where $\omega[n] \rightarrow$ window function, $x[n] \rightarrow$ given signal, $m \rightarrow$ discrete value and $w \rightarrow$ continuous value.

2) Wavelet Transform

Wavelet Transform is the method adopted to overcome the shortcomings of STFT. WT uses window width that will decrease with frequency. The following are the types of WT:

a) Continuous WT

$$T(a, b) = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} x(t) \varphi^* \left(\frac{t-b}{a} \right) dt \quad (3)$$

Where, $x(t)$ → given signal, $\varphi(t)$ → mother wavelet, $\varphi^*(t)$ → complex conjugate of $\varphi(t)$, a → scale of the wavelet, b → location of the wavelet.

b) Discrete WT

Discrete WT can be defined as the transform that decomposes the original signal into wavelet basis functions that are mutually orthogonal. In this work, Daubechies wavelet (db6) transform has been applied which has the properties of asymmetry, orthogonality, and biorthogonality.

C. Feature Learning

An autoencoder is said to be a self-supervised feature learning model to learn useful features from the given input. An autoencoder basically has three layers, namely input layer or encoder, hidden layer, and finally, output layer or decoder. The requirements for building an autoencoder are as follows:

- Encoding method
- Decoding method
- Loss function

There are different variants of autoencoder namely, denoising autoencoder (DAE), variational autoencoder (VAE), sparse autoencoder, and SAE. In this work, SAE has been applied with the binary cross-entropy loss function, adadelta optimizer, and sigmoid activation functions.

D. Classification

The quality of the classification model can be evaluated using performance metrics. The performance metrics were performed using python. For each class, the performance metrics show F1-score, Precision, and Recall. Dataset has split into 50:50 (training : testing) and 70:30 (testing : validation). The classification target labels are given as 0, 1, and 2 which implies rest, real or imagined motion of the left first or both fists, and real or imagined motion of the right first or both feet respectively. In this work the following approaches have been used for classification:

1) ML approach using SVM classifier

SVM is a classical supervised ML classifier that has the efficiency of classifying both linear and non-linear data. It has a benchmark of achieving good accuracy with low power of computation. The classification error can be measured using precision, recall, specificity and sensitivity.

2) DL approach using CNN model

In deep learning, the CNN model is a classical method for classification which is widely used in image and signal processing. The CNN model architecture has 5 basic layers namely the input layer followed by the Convolutional layer, pooling layer, fully connected layer and output layer. In this work, the CNN model consists of the input layer, three convolutional layers, a max-pooling layer, a dropout layer, a flatten layer, two dense layer, and finally a softmax layer.

IV. RESULTS AND DISCUSSION

A. Performance Metrics

This section gives the performance metrics of ML approaches with WT and STFT, and the accuracy of DL approaches using CNN with WT and STFT for 100 epochs.

1) Machine Learning Approach

The Multi-class ML approaches to classify the MI tasks include OvO and OvR using the SVM classifier has been used. TABLE I. gives the accuracy, precision (truly positive percentage), recall (true positive rate), and f1 score for ML approaches. The overall accuracy of the OvO and OvR approach with STFT and WT is 51.56% respectively. Also, precision, recall, and f1-score of OvO and OvR approach with STFT are zero for classes 1 and 2. It could be due to the inefficiency of the classifier to predict any positive result. It is also seen that OvO and OvR approaches with WT shows accuracy of 77.5% and 78.69% respectively in the Multi-class MI task for the given MI dataset.

2) Deep Learning Approach

The Multi-class DL classification approach to classify the MI tasks has been performed using the CNN model. TABLE II, gives the accuracy and AUC (Area Under Curve) score of the CNN model to classify 3-class MI tasks. The CNN model accuracy and AUC score with WT are found to be 96.68% and 0.87 respectively for the given EEG dataset. The CNN model accuracy and AUC score with STFT are found to be 51.45% and 0.51 respectively for the given EEG dataset. The reason for producing higher accuracy in the case of using WT while transforming EEG signal could be due to the windowing technique.

TABLE II. OVERALL ACCURACY AND AUC SCORE OF DEEP LEARNING APPROACHES FOR MOTOR CLASSIFICATION

DL Approaches	CNN model with STFT	CNN model with WT
Accuracy (%) for 100 epochs	51.45	96.68
AUC score	0.51	0.87

TABLE I. PERFORMANCE METRICS OF MULTI-CLASS MACHINE LEARNING APPROACHES FOR CLASSIFICATION

ML Approaches	OvO approach with STFT			OvR approach with STFT			OvO approach with WT			OvR approach with WT		
	Overall Accuracy = 51.56%			Overall Accuracy = 51.56%			Overall Accuracy = 77.5%			Overall Accuracy = 78.69%		
Class	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.52	1	0.68	0.53	1	0.69	0.74	0.96	0.83	0.75	0.92	0.83
1	0	0	0	0	0	0	0.87	0.64	0.74	0.83	0.73	0.78
2	0	0	0	0	0	0	0.83	0.52	0.64	0.86	0.56	0.68

B. ROC

A Receiver Operating Characteristic (ROC) curve is used to visualize graphically the classification performance for all thresholds of classification. The graph is a plot between the True Positive Rate (TPR) vs False Positive Rate (FPR). Fig 2, 3, 4, and 5 shows the ROC curve of the multi-class classification approach for OvR with WT, OvO with WT, OvR with STFT, and OvO with STFT respectively. It was also found that the area under the ROC curve (AUC_ROC) in Fig 2, 3, 4, and 5 is 0.92, 0.624, 0.50, and 0.50 respectively. From the AUC_ROC, Precision, Recall, and F1-score value, the OvR ML approach with wavelet transformed EEG signal for the given dataset is found to be more efficient than the other ML approach to classify the MI tasks.

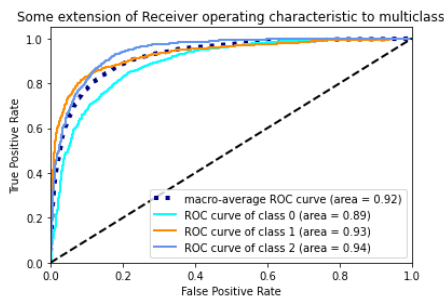


Fig. 2. ROC curve for OvR multi-class classification approach with WT.

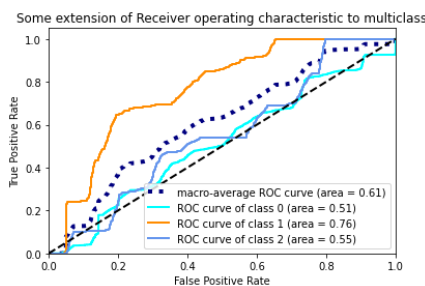


Fig. 3. ROC curve for OvO multi-class classification approach with WT

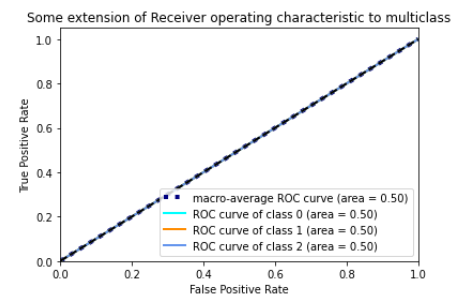


Fig. 4. ROC curve for OvR multi-class classification approach with STFT.

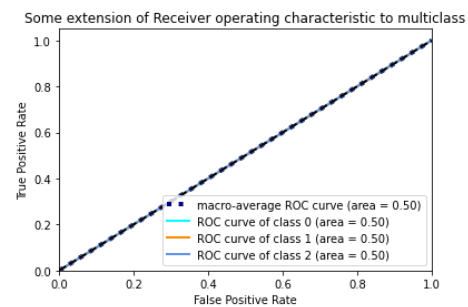


Fig. 5. ROC curve for OvO multi-class classification approach with STFT

C. Performance comparison

Fig 6 and 7 show the bar chart representation to compare the performance of different classification approaches (ML and DL) using Wavelet transform or Short-Time Fourier Transform. From Fig 6 it could be seen that for the given dataset DL model using CNN with WT outperforms the other two ML approaches (OvO and OvR) with WT. From Fig 7 it was observed that for the given dataset DL model, and ML approaches gives approximately the same classification accuracy of 51%.

1) Case 1

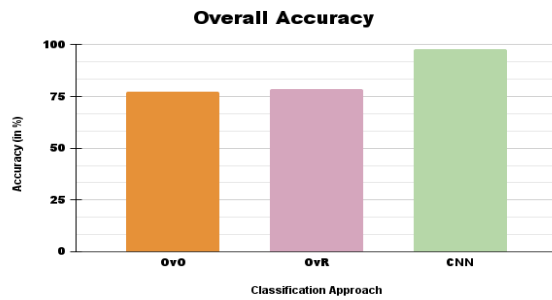


Fig. 6. Overall Accuracy of ML and DL approach with WT.

2) Case 2

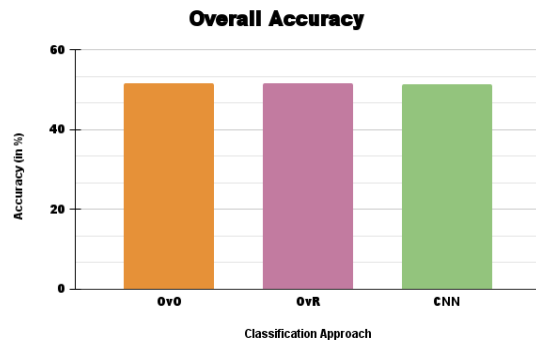


Fig. 7. Overall Accuracy of ML and DL approach with STFT.

D. Comparative analysis

In TABLE III, the comparative analysis of the proposed model with the existing model has been made. It can be seen that our proposed model has the highest accuracy of 96.68%.

TABLE III. COMPARATIVE ANALYSIS OF EXISTING METHOD WITH PROPOSED METHOD

S.No	MODEL	OVERALL ACCURACY
1.	MRC-MLP based feature extraction [11].	76%
2.	CWT based pre-processing followed by CNN for binary classification [12].	68.33%
3.	Stacked autoencoder based MI classification [14].	80.44%
4.	Deep-channel correlation network for 3-class MI classification [17].	87.03%
5.	CNN and stacked autoencoder for binary class MI classification [19].	82%
6.	Our proposed model.	96.68%

V. CONCLUSION

In this work, the performance comparison of ML and DL approaches to classify three-class MI EEG signals has been performed. Firstly, the MI dataset in which the EEG signals were recorded using 64 channels was analyzed in the time-frequency domain using either Wavelet Transform or Short-Time Fourier Transform. Then the transformed data is given as input to SAE for feature learning. Later, ML multi-class classification approach like OvO and OvR using SVM classifier and the DL classification approach

using CNN model was applied to classify MI tasks. It was observed that the CNN model with WT outperforms when compared to the CNN model with STFT and other ML approaches for the EEG-MI dataset used.

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